

Chapter 25 Capacitance

Capacitors and Capacitance

A capacitor is a device that stores electrical charge and energy, converting other forms of energy into electrical energy.

Two-conductor capacitor is composed of two conductors/electrodes separated by a certain medium or vacuum. After charging, two conductors carry equal amount of opposite charges.

And Capacitance:

$$C = \frac{Q}{V}$$

It is definition. Capacitance is independent of Q and V . Q is the amount of charge on a single conductor. (SI unit: F)

Calculating the Capacitance

- Assume a charge q on the plates
- Finding the electric field $\vec{E}$ by Gauss's Law
- Calculate the potential difference V by $V = Ed(\text{uniform})$ or $V = \int E dr$
- Calculating C from $C = \frac{q}{V}$.

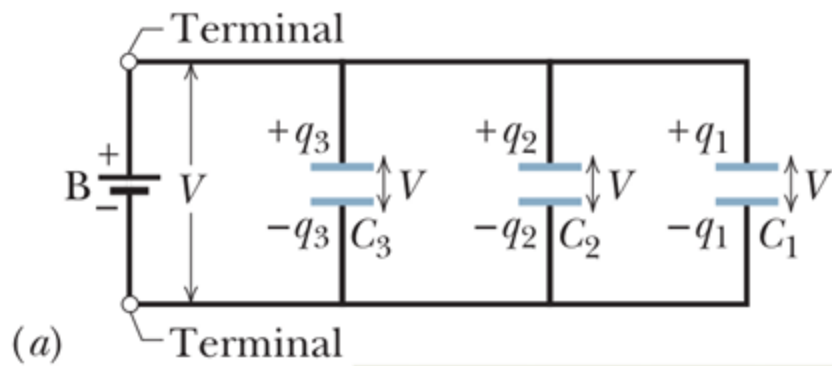
Capacitors in Series and Parallel

Capacitors in Parallel

$$Q = Q_1 + Q_2 + \dots$$

$$V = V_1 = V_2 = \dots$$

$$C_{eq} = C_1 + C_2 + \dots = \sum_{j=1}^n C_j$$



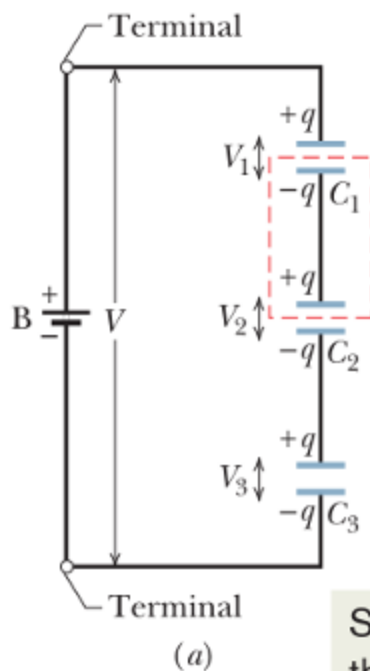
Parallel capacitors and their equivalent have the same V ("par-V").

Capacitors in Series

$$Q = Q_1 = Q_2 = \dots$$

$$V = V_1 + V_2 + \dots$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$



Series capacitors and their equivalent have the same q ("seri-q").

Energy Stored in An Electric-Field

During charging process, external forces do work and converting other forms of energy into electrical energy. Is is the energy of electric field.

Energy Storage in Capacitors:

$$U = \frac{1}{2} CV^2 = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} QV$$

Energy Density of Electric Field

$$V = Ed$$

$$E = \frac{\sigma}{\epsilon_0}$$

$$U = \frac{1}{2} QV = \frac{1}{2} \sigma AEd = \frac{1}{2} \epsilon_0 E^2 Ad$$

Thus, the energy density is

$$u = \frac{U}{Ad} = \frac{1}{2} \epsilon_0 E^2$$

Capacitor with A Dielectric

Dielectric are **insulating materials**. The functions of dielectric:

- Bring two conductors of a capacitor very close together without them touching. **Increases** the **capacitance** of capacitor.
- Limit the potential difference to certain value $V_{\max}$, called **breakdown potential**.

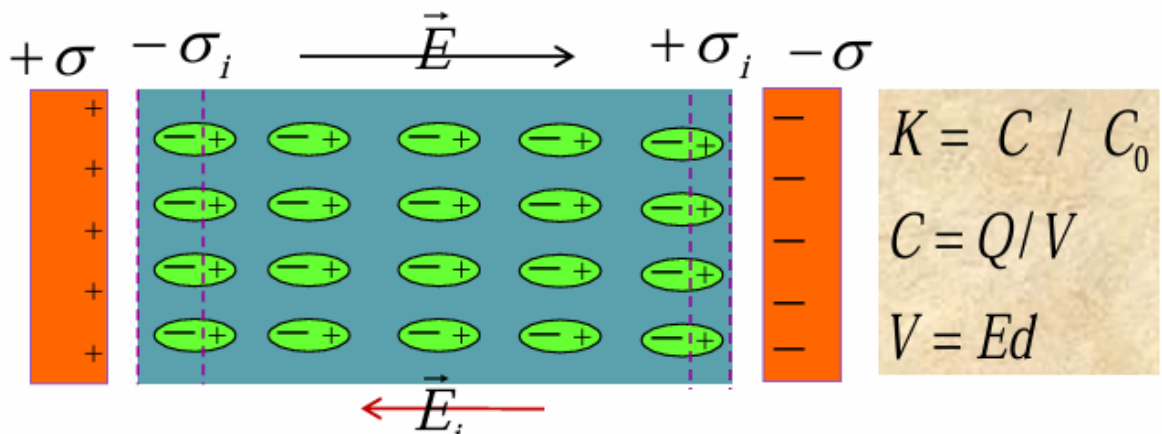
Dielectric constant:

$$\kappa = \frac{C}{C_0} = \frac{\epsilon}{\epsilon_0}$$

where C and ϵ is the **capacitance** and **permittivity** of capacitors containing dielectric. C_0 is the capacitance of vacuum.

Polarization of Dielectric

The net bound **charge is generated inside and on the surface of the dielectric** redistributed by an electric field. The charge is called **polarization charge(induced charge)** and this phenomenon is called **dielectric polarization**.



The electric field generated by the polarization charge is opposite in direction to original electric field.

The density of polarization charge and net polarization charge:

$$\sigma_i = -\sigma \left(1 - \frac{1}{\kappa}\right)$$
$$Q_i = -Q \left(1 - \frac{1}{\kappa}\right)$$

Dielectric breakdown: Under a sufficiently strong electric field, electrons are pulled out and collide with other molecules, releasing even more electrons. This instantaneous avalanche of electrons creates a spark/arc discharge, causing a short circuit phenomenon.